



LAB ASSIGNMENT -2
OPERATING SYSTEM (MCA -103)

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1. Introduction of internal and external commands.

- Internal Commands

Internal commands are built into the **shell** itself, so they do not exist as separate executable files.

Examples:

- cd
- echo
- pwd
- exit

```
csed@Ubuntu1:~/Desktop$ pwd
/home/csed/Desktop
csed@Ubuntu1:~/Desktop$ ls
folder1
csed@Ubuntu1:~/Desktop$ cd folder1
csed@Ubuntu1:~/Desktop/folder1$ pwd
/home/csed/Desktop/folder1
csed@Ubuntu1:~/Desktop/folder1$ cd ..
csed@Ubuntu1:~/Desktop$ ls
folder1
```

- External Commands:

External commands are separate executable programs stored in locations such as /bin or /usr/bin.

Examples:

- ls
- cat
- date
- mkdir
- grep

```
csed@Ubuntu1:~/Desktop$ echo "Welcome to the operating system lab"
Welcome to the operating system lab
csed@Ubuntu1:~/Desktop$ mkdir text && cd text
csed@Ubuntu1:~/Desktop/text$ mkdir text1.txt text2.txt text3.txt
csed@Ubuntu1:~/Desktop/text$ ls
text1.txt  text2.txt  text3.txt
csed@Ubuntu1:~/Desktop/text$
```

1. Feeding output of one command to another command by pipelining.

A **pipe** | sends the output of one command directly as input to another command.

Syntax

command1 | command2

```
csed@Ubuntu1:~$ ls -l | more
total 24
-rw-rw-r-- 1 csed csed  0 May 20 10:10 file1.txt
-rw-rw-r-- 1 csed csed  0 May 20 10:10 file2.txt
-rw-rw-r-- 1 csed csed  0 May 20 10:10 file3.txt
-rw-rw-r-- 1 csed csed  0 May 20 10:10 notes.txt
```

```
csed@Ubuntu1:~/Desktop$ mkdir text && cd text
csed@Ubuntu1:~/Desktop/text$ mkdir text1.txt text2.txt text3.txt
```

1. expr and Locating Commands

expr command

The expr command is used to perform **arithmetic and string operations**.

```
csed@Ubuntu1:~/Desktop$ expr 10 + 20
30
csed@Ubuntu1:~/Desktop$ expr 30 * 3
90
csed@Ubuntu1:~/Desktop$
```

Locating Commands

which

Shows the location of an executable command.

```
csed@Ubuntu1:~/Desktop$ which ls
/usr/bin/ls
csed@Ubuntu1:~/Desktop$
```

whereis

Locates the binary, source, and manual page of a command.

```
csed@Ubuntu1:~$ whereis ls
ls: /usr/bin/ls /usr/share/man/man1/ls.1.gz
```

locate

Searches for files using a database.

```
csed@Ubuntu1:~$ locate passwd
/etc/passwd
csed@Ubuntu1:~$
```

1. echo Command

The echo command displays text or the value of variables on the terminal.

```
csed@Ubuntu1:~/Desktop$ echo "Welcome to the operating system lab"
Welcome to the operating system lab
```

1. Using. And ..

In Linux directories:

- . means **current directory**
- .. means **parent directory**

```
csed@Ubuntu1:~/Desktop/folder1$ cd ..
csed@Ubuntu1:~/Desktop$ ls
folder1
csed@Ubuntu1:~/Desktop$ cd .
csed@Ubuntu1:~/Desktop$ pwd
/home/csed/Desktop
```

1. Ways of Signing Off from Linux

You can log out or end a Linux shell session using:

Method 1

exit

Method 2

logout

Method 3

Press:

Ctrl + D

```
csed@Ubuntu1:~$ exit
logout
Connection to localhost closed.
csed@Ubuntu1:~$ logout
Connection to localhost closed.
csed@Ubuntu1:~$
# or press Ctrl + D
Connection to localhost closed.
csed@Ubuntu1:~$
```

If using a graphical Linux desktop, you can also use the **Log Out/Sign Out option** from the system menu.

1. Ping, Man and help command.

- Ping

The ping command checks connectivity between your computer and another host

```

csed@Ubuntu1:~/Desktop$ ping google.com
PING google.com (142.250.193.14) 56(84) bytes of data.
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=1 ttl=115 time=14.5 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=2 ttl=115 time=14.2 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=3 ttl=115 time=13.7 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=4 ttl=115 time=14.0 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=5 ttl=115 time=14.3 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=6 ttl=115 time=13.8 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=7 ttl=115 time=13.8 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=8 ttl=115 time=13.8 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=9 ttl=115 time=14.3 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=10 ttl=115 time=17.4 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=11 ttl=115 time=14.1 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=12 ttl=115 time=14.1 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=13 ttl=115 time=13.9 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=14 ttl=115 time=13.9 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=15 ttl=115 time=14.0 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=16 ttl=115 time=14.1 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=17 ttl=115 time=14.0 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=18 ttl=115 time=14.1 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=19 ttl=115 time=16.7 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=20 ttl=115 time=13.8 ms
64 bytes from tzdelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=21 ttl=115 time=14.2 ms
^C
--- google.com ping statistics ---
21 packets transmitted, 21 received, 0% packet loss, time 26099ms
rtt min/avg/max/mdev = 13.738/14.315/17.352/0.898 ms

```

- **Man**

man means **manual**. It displays detailed documentation about Linux commands.

- **help**

The help command provides information about **shell built-in commands**.

```

csed@Ubuntu1:~$ man ls
csed@Ubuntu1:~$ help cd
cd: cd [-L | [-P [-e]] [-@]] [dir]
    Change the shell working directory.

```

1. Combining Commands

Linux allows commands to be combined in different ways.

Using ;

Runs commands one after another:

Using &&

The second command runs only if the first command succeeds.

Using ||

The second command runs if the first command fails.

Using a Pipe |

Passes output from one command to another.

```
csed@Ubuntu1:~$ pwd ; ls ; date
/home/csed
Desktop  Documents  Downloads  file.txt  Music  Pictures  Public  Templates  Videos
Mon May 20 10:20:30 IST 2024
csed@Ubuntu1:~$ mkdir test && cd test && touch a.txt && ls
a.txt
csed@Ubuntu1:~$ cd wrongdir || echo "Directory not found"
Directory not found
csed@Ubuntu1:~$ ls -l | grep "txt"
-rw-rw-r-- 1 csed csed  0 May 20 10:10 a.txt
-rw-rw-r-- 1 csed csed  0 May 20 10:10 file.txt
csed@Ubuntu1:~$
```

1. File Permissions and changing the access rights (chmod)

Linux has three basic permissions:

- r = Read
- w = Write
- x = Execute

Permissions apply to:

- User/Owner (u)
- Group (g)
- Others (o)

```
csed@Ubuntu1:~$ ls -l file.txt
-rw-rw-r-- 1 csed csed 0 May 20 10:10 file.txt
csed@Ubuntu1:~$ chmod u+x file.txt
csed@Ubuntu1:~$ ls -l file.txt
-rwxrw-r-- 1 csed csed 0 May 20 10:10 file.txt
csed@Ubuntu1:~$ chmod g-w file.txt
csed@Ubuntu1:~$ ls -l file.txt
-rwxr--r-- 1 csed csed 0 May 20 10:10 file.txt
csed@Ubuntu1:~$ chmod 755 file.txt
csed@Ubuntu1:~$ ls -l file.txt
-rwxr-xr-x 1 csed csed 0 May 20 10:10 file.txt
csed@Ubuntu1:~$
```

1. vi editor and its basics: write a small paragraph using vi editor.

The vi editor is a text editor available in Linux.

Create/Open a File(example)

vi paragraph.txt

Important Modes

1. Command Mode

This is the default mode. You can save, quit, delete, and perform other commands.

Press:

Esc

to enter command mode.

2. Insert Mode

Used to type or edit text.

Press:

I

```
csed@Ubuntu1:~$ vi paragraph.txt
( Press i to enter insert mode )
Linux is a powerful and widely used operating system.
It is secure, stable and multi-user.
Linux supports many applications and is open source.
It is used in servers, desktops, and mobile devices.
( Press Esc to exit insert mode )
( Type :wq and press Enter to save and quit )
```

```
csed@Ubuntu1:~$ cat paragraph.txt
Linux is a powerful and widely used operating system.
It is secure, stable and multi-user.
Linux supports many applications and is open source.
It is used in servers, desktops, and mobile devices.
csed@Ubuntu1:~$
```

Basic vi Commands

Command	Function
i	Insert text
Esc	Return to command mode
:w	Save file
:q	Quit
:wq	Save and quit
:q!	Quit without saving
dd	Delete current line
yy	Copy current line
p	Paste
/word	Search for a word